

Input:

- V . Set of nodes
- E_s . Set of scheduled edges (v, u, dep, arr)
- E_t . Set of transfer edges (v, w, Δ)
- $country(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets
- $bucket(t)$. Time bucket mapping function

Output:

- P . Sequence of labels representing a route within 24 hours

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1: Initialize  $Labels(v) \leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize queue  $Q \leftarrow \emptyset$ 
3: Initialize  $best\_label \leftarrow \text{null}$ 
4: // Initialize labels at departures
5: for all  $v \in V$  do
6:   for all scheduled departure from  $v$  at time  $dep$  with arrival  $arr$  do
7:      $L \leftarrow (v, dep, \{country(v)\}, \text{pred} = \text{null}, \text{start\_time} = dep)$ 
8:      $Labels(v) \leftarrow Labels(v) \cup \{L\}$ 
9:      $Q \leftarrow Q \cup \{L\}$ 
10: // Main loop
11: while  $Q \neq \emptyset$  do
12:   extract  $L = (v, t, C, \text{pred}, \text{start\_time})$  from  $Q$ 
13:   // Enforce 24-hour journey limit
14:   if  $t - \text{start\_time} > 24 : 00$  then
15:     continue
16:   // Upper bound pruning
17:    $b \leftarrow bucket(t)$ 
18:    $UB \leftarrow |R(v, b) \setminus C|$ 
19:   if  $best\_label \neq \text{null} \wedge |C| + UB \leq |best\_label.C|$  then
20:     continue
21:   // Update best label
22:   if  $best\_label = \text{null} \vee |C| > |best\_label.C|$  then
23:      $best\_label \leftarrow L$ 
24:   // Extend via scheduled edges
25:   for all  $(v, u, dep, arr) \in E_s$  do
26:     if  $dep \geq t \wedge dep - \text{start\_time} \leq 24 : 00$  then
27:        $t_{\text{new}} \leftarrow arr$ 
28:       if  $t_{\text{new}} - \text{start\_time} > 24 : 00$  then
29:         continue
30:        $C' \leftarrow C \cup \{country(u)\}$ 
31:        $L' \leftarrow (u, t_{\text{new}}, C', \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
32:       if  $\nexists L'' \in Labels(u)$  such that
33:          $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C' \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C'))$  then
34:           remove all  $L'' \in Labels(u)$  such that
35:              $(t_{\text{new}} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C' \neq L''.C))$ 
36:            $Labels(u) \leftarrow Labels(u) \cup \{L'\}$ 

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35:   |   |   |    $Q \leftarrow Q \cup \{L'\}$ 
36:   // Extend via transfers
37:   for all  $(v, w, \Delta) \in E_t$  do
38:       |    $t_{\text{new}} \leftarrow t + \Delta$ 
39:       |   if  $t_{\text{new}} - \text{start\_time} > 24 : 00$  then
40:           |   continue
41:       |    $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
42:       |   if  $\nexists L'' \in \text{Labels}(w)$  such that
43:           |        $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$  then
44:               |       remove all  $L'' \in \text{Labels}(w)$  such that
45:                   |            $(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$ 
46:                   |            $\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$ 
47:                   |    $Q \leftarrow Q \cup \{L'\}$ 
48: // Reconstruct path
49:  $P \leftarrow$  empty sequence
50:  $L \leftarrow \text{best\_label}$ 
51: while  $L \neq \text{null}$  do
52:     |   prepend  $L$  to  $P$ 
53:     |    $L \leftarrow L.\text{pred}$ 
54: return  $P$ 

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